

Lab assignments for Git-GitHub

1. Create a new Git repository and initialize it using `git init`. Check the repository status.

```
workspace@ubuntu:~/asgnmt/git-assignment$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/workspace/asgnmt/git-assignment/.git/
workspace@ubuntu:~/asgnmt/git-assignment$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
workspace@ubuntu:~/asgnmt/git-assignment$
```

2. Create a file named `README.md`, add some content, and commit it with an appropriate commit message.

```
nothing to commit (create/copy files and use "git add" to track)
workspace@ubuntu:~/asgnmt/git-assignment$ echo "# Git Assignment" > README.md
workspace@ubuntu:~/asgnmt/git-assignment$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md

nothing added to commit but untracked files present (use "git add" to track)
workspace@ubuntu:~/asgnmt/git-assignment$ git add .
workspace@ubuntu:~/asgnmt/git-assignment$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md

workspace@ubuntu:~/asgnmt/git-assignment$ git commit -m "ADD README file"
[master (root-commit) 17e7352] ADD README file
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
workspace@ubuntu:~/asgnmt/git-assignment$ git log
commit 17e73529133510fd466f9b814fd6876d7fd1a247 (HEAD -> master)
Author: Praanshu <pranshusahu.pranshu@gmail.com>
Date:   Mon Sep 14 06:46:19 2026 +0530

    ADD README file
```

3. Create a new branch named `feature`. Switch to this branch and make a change to `README.md`.

```
workspace@ubuntu:~/asgnmt/git-assignment$ git checkout -b feature
Switched to a new branch 'feature'
workspace@ubuntu:~/asgnmt/git-assignment$ git branch
* feature
  master
workspace@ubuntu:~/asgnmt/git-assignment$ echo "Edited by feature branch" >> README.md
workspace@ubuntu:~/asgnmt/git-assignment$ git status
On branch feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
workspace@ubuntu:~/asgnmt/git-assignment$
```

4. Create a second file, add it to the staging area, and verify the staged changes using `git status` and `git diff --staged`.

```
workspace@ubuntu:~/asgnmt/git-assignment$ echo "This is the second file." >second.txt
workspace@ubuntu:~/asgnmt/git-assignment$ git add second.txt
workspace@ubuntu:~/asgnmt/git-assignment$ git status
On branch feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   second.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

workspace@ubuntu:~/asgnmt/git-assignment$ git diff --staged
diff --git a/second.txt b/second.txt
new file mode 100644
index 0000000..578b181
--- /dev/null
+++ b/second.txt
@@ -0,0 +1 @@
+This is the second file.
workspace@ubuntu:~/asgnmt/git-assignment$
```

5. Make two separate commits with meaningful commit messages. Display the commit history using `git log`.

```
workspace@ubuntu:~/asgmt/git-assignment$ git add README.md
workspace@ubuntu:~/asgmt/git-assignment$ git commit -m "Update README on feature branch"
[feature e8a0669] Update README on feature branch
 2 files changed, 2 insertions(+)
 create mode 100644 second.txt
workspace@ubuntu:~/asgmt/git-assignment$ git add second.txt
workspace@ubuntu:~/asgmt/git-assignment$ git commit -m "Add second file"
On branch feature
nothing to commit, working tree clean
workspace@ubuntu:~/asgmt/git-assignment$ git log
commit e8a0669e5fa5d01e9f5a168e2403584869bc235e (HEAD -> feature)
Author: Praanshuu <pranshusahu.pranshu@gmail.com>
Date: Mon Sep 14 06:52:46 2026 +0530

    Update README on feature branch

commit 17e73529133510fd466f9b814fd6876d7fd1a247 (master)
Author: Praanshuu <pranshusahu.pranshu@gmail.com>
Date: Mon Sep 14 06:46:19 2026 +0530

    ADD README file
```

6. Rename the current branch to `main` and verify the current branch.

```
workspace@ubuntu:~/asgmt/git-assignment$ git branch
* feature
  master
workspace@ubuntu:~/asgmt/git-assignment$ git branch -M main
workspace@ubuntu:~/asgmt/git-assignment$ git branch
* main
  master
```

7. Create a GitHub repository and connect the local Git repository to it using a remote named `origin`.

```
workspace@ubuntu:~/asgmt/git-assignment$ git remote add origin https://github.com/Praanshuu/git-assignment.git
workspace@ubuntu:~/asgmt/git-assignment$ git remote -v
origin  https://github.com/Praanshuu/git-assignment.git (fetch)
origin  https://github.com/Praanshuu/git-assignment.git (push)
```

8. Push the local `main` branch to the GitHub repository and verify that the files appear on GitHub.

```
fatal: Authentication failed for 'https://github.com/Praanshoo/git-assignment.git/'
workspace@ubuntu:~/asgmt/git-assignment$ git push -u origin main
Username for 'https://github.com': Praanshoo
Password for 'https://Praanshoo@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (7/7), 583 bytes | 145.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Praanshoo/git-assignment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
workspace@ubuntu:~/asgmt/git-assignment$
```

The screenshot shows the GitHub interface for a repository named 'git-assignment' by user 'Praanshoo'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows two commits by 'Praanshoo' on the 'main' branch, both titled 'Update README on feature branch', with the most recent commit being 'e8a0669' from 7 minutes ago. The file list shows 'README.md' and 'second.txt', both updated 7 minutes ago. The README content is visible, showing the title 'Git Assignment' and the text 'Edited by feature branch'.

git-assignment Public

main 1 Branch 0 Tags

Go to file T Add file <> Code

Praanshoo Update README on feature branch e8a0669 · 7 minutes ago 2 Commits

File	Commit	Time
README.md	Update README on feature branch	7 minutes ago
second.txt	Update README on feature branch	7 minutes ago

README

Git Assignment

Edited by feature branch

9. Create a new branch named `dev`, make a change, commit it, and push the branch to GitHub.

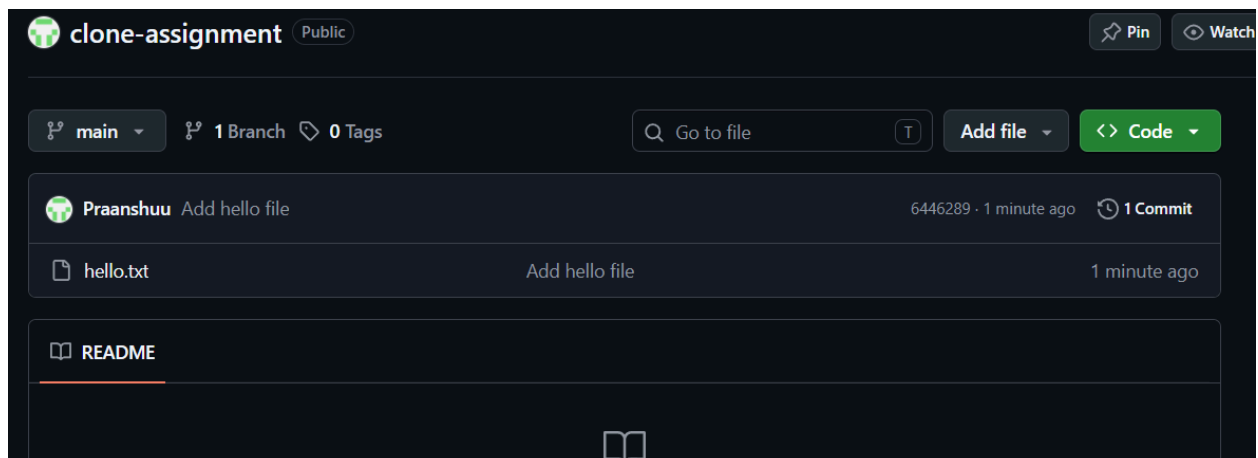
```
workspace@ubuntu:~/asgmt/git-assignment$ git checkout -b dev
Switched to a new branch 'dev'
workspace@ubuntu:~/asgmt/git-assignment$ echo "Changes made in dev branch." >> README.md
workspace@ubuntu:~/asgmt/git-assignment$ git push -u origin dev
Username for 'https://github.com': Praanshuu
Password for 'https://Praanshuu@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 367 bytes | 367.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/Praanshuu/git-assignment/pull/new/dev
remote:
To https://github.com/Praanshuu/git-assignment.git
 * [new branch]      dev -> dev
branch 'dev' set up to track 'origin/dev'.
workspace@ubuntu:~/asgmt/git-assignment$ git branch
* dev
  main
  master
workspace@ubuntu:~/asgmt/git-assignment$
```

10. Switch back to `main` and merge the `dev` branch into it. Push the updated `main` branch to GitHub.

```
workspace@ubuntu:~/asgmt/git-assignment$ git checkout main
M README.md
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
workspace@ubuntu:~/asgmt/git-assignment$ git merge dev
Already up to date.
workspace@ubuntu:~/asgmt/git-assignment$ git push origin main
Username for 'https://github.com': Praanshuu
Password for 'https://Praanshuu@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Praanshuu/git-assignment.git
 e8a0669..f5cf4d1  main -> main
workspace@ubuntu:~/asgmt/git-assignment$
```

11. Create a GitHub repository, clone it to your computer using `git clone`, add a file, commit the change, and push it back to GitHub.

```
workspace@ubuntu:~/asgnmt$ git clone https://github.com/Praanshoo/clone-assignment.git
Cloning into 'clone-assignment'...
warning: You appear to have cloned an empty repository.
workspace@ubuntu:~/asgnmt$ cd clone-assignment
workspace@ubuntu:~/asgnmt/clone-assignment$ echo "Hello GitHub"
Hello GitHub
workspace@ubuntu:~/asgnmt/clone-assignment$ echo "Hello GitHub" > hello.txt
workspace@ubuntu:~/asgnmt/clone-assignment$ git add hello.txt
workspace@ubuntu:~/asgnmt/clone-assignment$ git commit -m "Add hello file"
[main (root-commit) 6446289] Add hello file
 1 file changed, 1 insertion(+)
 create mode 100644 hello.txt
workspace@ubuntu:~/asgnmt/clone-assignment$ git push
Username for 'https://github.com': Praanshoo
Password for 'https://Praanshoo@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Praanshoo/clone-assignment.git/'
workspace@ubuntu:~/asgnmt/clone-assignment$ git push
Username for 'https://github.com': Praanshoo
Password for 'https://Praanshoo@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 231 bytes | 115.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Praanshoo/clone-assignment.git
 * [new branch]      main -> main
workspace@ubuntu:~/asgnmt/clone-assignment$
```



12. Use `git remote -v` to display the remote repository URL. Change the remote URL to another GitHub repository and verify the change.

```
workspace@ubuntu:~/asgmt/clone-assignment$ git remote -v
origin  https://github.com/Praanshuu/clone-assignment.git (fetch)
origin  https://github.com/Praanshuu/clone-assignment.git (push)
workspace@ubuntu:~/asgmt/clone-assignment$ git remote set-url origin https://github.c
om/Praanshuu/clone-assignment.git
workspace@ubuntu:~/asgmt/clone-assignment$ git remote -v
origin  https://github.com/Praanshuu/clone-assignment.git (fetch)
origin  https://github.com/Praanshuu/clone-assignment.git (push)
workspace@ubuntu:~/asgmt/clone-assignment$
```

13. Make a change to a tracked file and use `git diff` to view the changes before committing them.

```
workspace@ubuntu:~/asgmt/clone-assignment$ echo "Another change" >> hello.txt
workspace@ubuntu:~/asgmt/clone-assignment$ git diff
diff --git a/hello.txt b/hello.txt
index bca999f..9456cbd 100644
--- a/hello.txt
+++ b/hello.txt
@@ -1,2 @@
   Hello GitHub
+Another change
```

14. Undo the most recent commit while keeping the changes in the working directory. Verify the result using `git status` and `git log`.

```
workspace@ubuntu:~/asgmt/clone-assignment$ git add hello.txt
workspace@ubuntu:~/asgmt/clone-assignment$ git commit -m "Add another change"
[main 77c9cbb] Add another change
1 file changed, 1 insertion(+)
workspace@ubuntu:~/asgmt/clone-assignment$ git reset --soft HEAD-1
fatal: ambiguous argument 'HEAD-1': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
workspace@ubuntu:~/asgmt/clone-assignment$ git reset --soft HEAD~1
workspace@ubuntu:~/asgmt/clone-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   hello.txt

workspace@ubuntu:~/asgmt/clone-assignment$ git log
commit 6446289935ae4934ab9eba726c9cdd8f2b27872c (HEAD -> main, origin/main)
Author: Praanshuu <pranshusahu.pranshu@gmail.com>
Date:   Mon Sep 14 07:08:55 2026 +0530

    Add hello file
```

15. Create a `.gitignore` file that ignores files with the `.log` extension and a directory named `temp`. Verify that these files are not tracked by Git.

```
workspace@ubuntu:~/asgmt/clone-assignment$ echo "*.log" > .gitignore
workspace@ubuntu:~/asgmt/clone-assignment$ echo "temp/" >> .gitignore
workspace@ubuntu:~/asgmt/clone-assignment$ cat .gitignore
*.log
temp/
workspace@ubuntu:~/asgmt/clone-assignment$ echo "some log" > test.log
workspace@ubuntu:~/asgmt/clone-assignment$ mkdir temp
workspace@ubuntu:~/asgmt/clone-assignment$ echo "temp file" > temp/file.txt
workspace@ubuntu:~/asgmt/clone-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   hello.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

workspace@ubuntu:~/asgmt/clone-assignment$ git add .gitignore
workspace@ubuntu:~/asgmt/clone-assignment$ git commit -m "Add gitignore rules"
[main 2af21c0] Add gitignore rules
 2 files changed, 3 insertions(+)
 create mode 100644 .gitignore
workspace@ubuntu:~/asgmt/clone-assignment$
```